

Oluwatobi Melvyn Mayungbo

AI Engineer, LLM and Retrieval Systems

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SUMMARY

AI engineer who builds agent tools and the checks around them. Built an MCP server of read-only tools for AI assistants, a retrieval assistant with guardrails, golden-set evaluation and citation verification, and a ledger API with idempotency, audit logs and reversals instead of edits. Works in Python, PostgreSQL and Docker.

SKILLS

AI and LLM Systems: Model Context Protocol (MCP), LLM Evaluation and Guardrails, Prompt Engineering, Retrieval-Augmented Generation (RAG), Citation Verification, Hybrid Search (BM25, Dense Embeddings, RRF), Hugging Face, ONNX Runtime

Engineering and MLOps: Python, PostgreSQL, FastAPI, Docker, CI/CD (GitHub Actions), SQL, DuckDB, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Pytest, Streamlit, React

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

Payments and Risk: Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

PROJECTS

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.
- Designed a rule-based, fully explainable scorer where findings carry stable IDs and decisive findings set a floor of 75 so trust signals can never hide them; each score carries a confidence level based on how many sources answered.
- Built a resilient HTTP layer with caching, per-source rate limits and retries with exponential backoff and jitter; one failing source never breaks an investigation, and missing data lowers confidence rather than the score.
- Assembled an evaluation set of 34 hand-checked addresses (12 risky, 22 safe) with a record and replay harness reporting ROC AUC, precision and recall.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Designed generation to fail safely under uncertainty: a sentinel-token refusal mechanism enforced across the prompt, output guardrail and evaluation harness, plus post-generation citation verification that strips any marker the model cited but never retrieved.
- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.
- Deployed on GCP Cloud Run via Docker within a ~225MB peak memory footprint (2GB budget), with per-session and daily request caps in application code to bound API cost on a public endpoint.

Production-Grade Double-Entry Ledger API | [GitHub](#)

Tech: Python 3.12, FastAPI, PostgreSQL 16, Redis 7, Alembic, Docker

- Built a double-entry ledger and payments API where correctness under concurrency, retries, partial failure and replay is enforced in the database: integer minor units, every transaction summing to zero through a deferred constraint trigger, and an append-only ledger where corrections are reversals.

- Required an Idempotency-Key on every mutating endpoint so a replayed key returns the original response and creates nothing; events go out through a transactional outbox.
- Wrote 138 tests against real PostgreSQL and Redis, including a concurrency test where exactly half of N simultaneous transfers succeed and no balance goes negative.

Credit Risk Decisioning and Fairness Audit System | [GitHub](#) | [Live App](#)

Tech: Python, LightGBM, scikit-learn, SHAP, WOE Scorecards, FastAPI, Docker

- Developed a full credit decisioning system producing calibrated default probabilities, cost-optimal approve and decline cutoffs, adverse action reason codes and an append-only audit trail that lets any individual decision be reconstructed months later.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action